

**PREPARATION AND QUALITY ASSESSMENT OF JACKFRUIT**  
**(*Artocarpus heterophyllus* Lam.) SEED HUMMUS**

*Project Report submitted in partial fulfilment of requirements*

*for the award of*

**Bachelor of Science in FOOD TECHNOLOGY**

*of*

**Hindustan Institute of Technology and Science**

*By*

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**HINDUSTAN**  
**INSTITUTE OF TECHNOLOGY & SCIENCE**  
**(DEEMED TO BE UNIVERSITY)**  
————— CHENNAI —————

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**Department of Food Technology,**  
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# HINDUSTAN

INSTITUTE OF TECHNOLOGY & SCIENCE  
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CHENNAI

Padur, Kelambakkam, Chennai-603103, Tamil Nadu

## DEPARTMENT OF FOOD TECHNOLOGY

### BONAFIDE CERTIFICATE

*This is to certify that the project report entitled*

**PREPARATION AND QUALITY ASSESSMENT OF JACKFRUIT ( *Artocarpus heterophyllus* Lam.) SEED HUMMUS**

*Submitted by*

**ANKITA TIRKEY (0019349024)**

In partial fulfilment of the requirements for the award of the Degree of Bachelor of Science in **Food Technology of Hindustan Institute of Technology and Science**, is a bonafide account of the work carried out by the above students of this department during the academic year 2021-2022 under our supervision.

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## **DECLARATION**

I hereby declare that the thesis entitled “**PREPARATION AND QUALITY ASSESSMENT OF JACKFRUIT (*Artocarpus heterophyllus* Lam.) SEED HUMMUS**” is a partial submission to the Hindustan Institute of Technology and Science for the award of degree of Bachelor of Science in Food Technology is my own work carried out under the Guidance and Supervision of Dr. Samuel Ayofemi Olalekan Adeyeye, Associate Professor, Department of Food Technology, Hindustan Institute of Technology and Science, Chennai, and that to the best of my knowledge and belief, it contains no material previously published or written by another person or material which has been accepted for the award of any other degree or diploma of the University or other institute or university of higher learning.

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## ABSTRACT

The study aimed at substituting jackfruit seed in place of chickpea in hummus. Jackfruit seeds were boiled for 1 hour at 100° C in order to reduce its alkaloid and oxalate content. After boiling, the alkaloid and oxalate content were reported to be 1.05 and 0.05 mg/100g respectively. Boiled jackfruit seeds were used for hummus production and jackfruit seed hummus samples were evaluated for nutritional (energy, moisture, total ash, carbohydrate, protein, fat, and fibre) value, and total flavonoid content. Results of nutritional value analysis showed that energy, moisture, total ash, carbohydrate, protein, fat, and fibre content per 100 g were 329.4 Kcal, 18.36, 0.82, 74.59, 4.79, 1.32, and 0.12 grams respectively. The value of total flavonoid in the formulated hummus was found to be 1.16 mg/100g. The sensory evaluation results of the hummus concluded with a decently accepted score of 8 out of 9 in terms of taste, and not less than 7.5 out of 9 in terms of other sensory attributes. The results showed that boiling reduced the alkaloid and oxalate content of the raw seeds changed from 8.77 - 8.93 to 1.05, and 5.47 - 5.59 to 0.05 mg/100g respectively, making them fit for consumption. But the processing done for hummus formulation did not have a significant effect on the total flavonoid content of the raw seed; it changed from 1.97 - 2.09 to 1.16 mg/100g. In conclusion, jackfruit seed hummus emerged as a high-carb, low-fat food, making it one of the dietary treatments for obesity, and good for weight maintenance when combined with physical activity and behavioral strategies.

**Keywords:** jackfruit seed hummus, high-carb and low-fat food, obesity, weight maintenance, jackfruit seed, jackfruit, jackfruit variety, jackfruit seed flour, traditional hummus, hummus, chickpea.

## CHAPTER 1 - INTRODUCTION

Jackfruit (*Artocarpus heterophyllus* Lam.), the world's largest tree-borne tropical fruit, is a member of the Moraceae family (Baliga *et al.*, 2011). The jackfruit is native to India, with its natural environment in the Western Ghats rainforest in the country's Southwest. It is currently a significant crop in India, Sri Lanka, Burma, China, Malaysia, Thailand, Indonesia and also the Philippines as stated by Haq (2006). Jackfruit can be consumed in both ripened and unripened forms. The jackfruit is the national fruit of Bangladesh and Sri Lanka, and state fruit of the Indian states of Karnataka, Kerala and Tamil Nadu. According to an article published on jackfruit by Kerala Agricultural University (2022) any increase in the water table or insufficient air of the soil is harmful to the crop. The largest fruit of the world 'jackfruit' is called by different names all over India - Chakka (Malayalam), Kathal (Hindi), Enchor (Bangla) and Halasu, Kujee, Hannu in (Kannada).

The fruit axis, the persistent perinanth and the true fruit are the three main regions a jackfruit is made up of, the perinanth is made of the bulb (the lower fleshy edible region), the middle-fused region that forms the rind of the syncarp, nonedible region widely known as the spikes (Prakash *et al.*, 2009). When ripe, the jackfruit's outer skin, which is constituted of hexagonal conical apex, turns greenish to brownish yellow. The pulp might be amber, dark yellow or orange in coloration. Jackfruit is a composite fruit containing seeds. The tree or fruit exudes a sticky, white latex when cut. Jackfruit consists of a skin, delicious yellow bulbs, fit for human consumption. Jackfruit seeds are light brown to brown in hue, rounded and 2-3 cm long by 1-1.5 cm wide. They are encased in a slender, whitish outer layer (Menaka *et al.*, 2011). As per Madruga *et al.* (2014), well-succulent, aromatic and delicious yellow bulb of jackfruit contain about 100 to 500 seeds, accounting for 8 to 15% of the overall fruit weight. The seeds are harvested from the mature fruit in various regions of South India, dried in the sun, and stored suitably for use during the rainy season. Jackfruit seeds are normally discarded as waste due to their perishable nature, however they have a one-month shelf life when maintained in a cool, moist setting.

The vitamin C in jackfruit plays a major role in the production of collagen, and also provides firmness to the skin. Potassium in jackfruit helps to lower blood pressure and reverse the effects of sodium, which raises blood pressure and affects the heart and blood vessels. As a result, heart diseases, strokes and bone loss are prevented and muscle, nerve functions are improved. Jackfruit is used as a meat substitute often. The unripe flesh has a texture similar to chicken or pulled pork

when cooked. For the most part, jackfruit is both safe and nutritious. Anyone allergic to latex or birch pollen, on the other hand, should refrain from consuming or handling the fruit (Lillis, 2019). Jackfruit is widely known for its anti-bacterial, anti-fungal, antidiabetic, anti-inflammatory and antioxidant activities (Shanmugapriya *et al.*, 2011). Jackfruit is high in phytonutrients including carotenoids, which can act as antioxidants (Baliga *et al.*, 2011). In addition, a high fibre content in jackfruit keeps bowel movements regular and reduces constipation and the seeds and flesh of jackfruit are said to be a cooling and nutritious tonic (Mushumbusi, 2015). Jackfruit is rich in key minerals such as magnesium which aids calcium absorption while also strengthening bones and preventing bone-related diseases like osteoporosis. Copper is crucial in thyroid gland metabolism and iron in jackfruit helps to prevent anaemia and aids in normal blood circulation.

Jackfruit has remained a minor tropical fruit in the market. Despite the fact that jackfruit is significant for a variety of goods no relevant research is conducted. Due to its short life and lack of processing facilities in the locations where it is grown, jackfruit is underutilised and not recognised as a commercial crop, and it is rarely planted on regular plantation scale (Reddy *et al.*, 2004). The inedible components of the jackfruit such as the outer prickly rind, inner perigone and centre core make up about 60% of the overall fruit (Huang *et al.*, 2018), whereas the edible flesh makes up only about 35% of the whole fruit (Narasimbham *et al.*, 1990). High-yielding jackfruit varieties, proper harvesting and post-harvest practises such as appropriate handling, transportation, storage, development of novel processing technologies and searching for new applications to minimise post-harvest and production losses, as well as conversion of jackfruit waste into value-added products would be better options for popularising jackfruit cultivation for consumption, along with waste management. Jackfruit flesh is perishable and flavour loss, tissue softening, cut base browning are common (Mondal *et al.*, 2013). The weakening of the fruit increases its vulnerability to bruising and mechanical damage (Ramli, 2009). Large numbers of ripe jackfruits deteriorate quickly after harvesting due to inadequate postharvest understanding resulting in poor management and insufficiency of hygienic practices and storage conditions in locations where they are produced and commercialised (Mondal *et al.*, 2013; Jagtap *et al.*, 2011). To combat the mechanical damage harvesting of jackfruit should be carried out while it is still green and mature. Adapting to proper post-harvest procedures may also make exporting easier by extending shelf life. The shelf life of whole jackfruit can be extended by two weeks when stored at 10°C and 85-90 percent humidity (Ramli, 2009).

Consumer desire for wholesome nutritious and convenient food products is on the rise right now. As a result, there has been a developing market for minimally processed fruits in recent years, raising concerns about preventing fruit damage through proper handling and storage methods (Torres *et al.*, 2017). Converting jackfruits into value-added goods, such as precut or ready-to-eat bulbs, may make it easier to separate the bulbs from the rind thus saving time allowing it to be commercialised among urban populations. Only a few recent studies have concentrated on prolonging the shelf life of jackfruit and adding value to jackfruit trash by converting it to various products and renewable energy sources. As a result, additional research should be focused on identifying potential industrial applications for jackfruit as well as proper waste management in the jackfruit processing industry.

The next main component of the study is hummus and hummus is thought to have originated in ancient Egypt, according to historical evidence. Based on numerous ancient records, hummus was first mentioned in Egypt in the 13th century. In the Middle Eastern region chickpeas are available abundantly and are consumed widely.

The Arabic word ‘hummus’ denotes chickpea. In the 13th century, historical documents describe a dish that was quite close to the hummus we consume today, and was enjoyed in Cairo. However, this does not prevent other regions from claiming hummus to be their own (Fayed, 2019). Cooked, mashed chickpeas are combined with tahini, olive oil, lemon juice, and spices to make traditional hummus. Other types of hummus, bean-based dips termed as hummus but not made according to the conventional hummus recipe are available in the market, each with its own set of components that may or may not make significant contributions to nutrient intakes or provide benefits beyond basic nutrition (Wallace *et al.*, 2016).

## **AIM AND OBJECTIVES**

Aim - To prepare jackfruit seed hummus and assess its quality parameters

Objectives-

- To check the alkaloid and oxalate content of boiled jackfruit seed
- To formulate the jackfruit seed hummus
- To check proximate nutrients and total flavonoids in the formulated hummus
- To evaluate its sensory attributes

## CHAPTER 2 - REVIEW OF LITERATURE

### 2.1. Characteristics of jackfruit varieties

Australia's cultivars include Black Gold, Cheena, Cochin, Golden Nugget, Honey Gold, and Lemon Gold. Chompa Gob is a well-known jackfruit variety from Singapore. Thailand's two varieties are Dang Rasimi and Kun Wi Chan. Malaysian cultivars include J-30, J-31, and NS1. Black Gold and Tabouey have the same shape (long and tapered), while Dang Rasimi and J-30 have the same shape (uniform and oblong), and Honey Gold, Lemon Gold, and NS1 have a blocky shape. The majority of jackfruit cultivars from various origins have a firm texture, and the ideal season for them is between the period of May-October (Table 2.1).

As per an article issued by the Kerala Agricultural University on jackfruit, Kerala is suggested to have two different categories of jackfruit - sugary, quite firm fleshed variety muttom varikka and sindoor which is high in sugar content with an appealing sunset orange coloured flakes, yields medium sized 11-12 kg of fruit twice a year.

According to the information gathered from Tamil Nadu Agricultural University Agritech portal the varieties of jackfruit found in Tamil Nadu are Velipala, Singapore varrika, Hybrid jack, Panruti selection, Thanjavur jack, Burliar1, PLR1, PLR(J)2 and PPI1. The mentioned jackfruit varieties are specifically found in Cuddalore, Kanyakumari, Dindigul, Pudukottai, Namakkal, Tirunelveli, Nilgiris district of Tamil Nadu.

**Table 2.1. Jackfruit varieties and its characteristics**

| Cultivar and origin    | Colour of the pulp | Texture of the pulp             | Fruit shape           | Quality                    | Season and month        |
|------------------------|--------------------|---------------------------------|-----------------------|----------------------------|-------------------------|
| Black Gold (Australia) | Deep orange        | Medium firm to soft             | Long, tapered         | Good, sweet, aromatic      | Late, September-october |
| Cheena (Australia)     | Deep orange        | Soft, melting, slightly fibrous | Long, narrow, uniform | Excellent, highly aromatic | Mid, July-August        |

|                              |                          |                            |                  |                                           |                             |
|------------------------------|--------------------------|----------------------------|------------------|-------------------------------------------|-----------------------------|
| Cochin<br>(Australia)        | Yellow to orange         | Firm to<br>crunchy         | Round, irregular | Good, mild                                | Early, June-July            |
| Chompa Gob<br>(Singapore)    | Orange                   | Firm                       | Blocky, uniform  | Good, mild<br>flavour                     | Mid, July-August            |
| Dang Rasimi<br>(Thailand)    | Deep orange              | Firm to soft               | Uniform, oblong  | Mild, sweet<br>flavour, pleasant<br>aroma | Mid, July-August            |
| Golden Nugget<br>(Australia) | Deep orange              | Soft to<br>medium firm     | Round            | Excellent<br>flavour                      | Early, May-June             |
| Honey Gold<br>(Australia)    | Dark yellow to<br>orange | Firm                       | Blocky           | Sweet rich<br>flavour and<br>aroma        | Mid, July-August            |
| J-30<br>(Malaysia)           | Deep orange              | Firm                       | Uniform, oblong  | Sweet rich<br>flavour and<br>aroma        | Mid, July-August            |
| J-31<br>(Malaysia)           | Deep yellow              | Firm                       | Irregular        | Sweet rich<br>flavour,<br>earthy aroma    | Early, May-June             |
| Kun Wi Chan<br>(Thailand)    | Yellow                   | Moderately<br>firm to soft | Uniform, round   | Mild, pleasant<br>flavour                 | Mid, July-August            |
| Lemon Gold<br>(Australia)    | Lemon yellow             | Firm                       | Blocky           | Sweet and<br>aromatic flavour             | Mid, July-August            |
| NS1<br>(Malaysia)            | Dark orange              | Firm                       | Blocky           | Sweet rich<br>flavour                     | Early, May-June             |
| Tabouey<br>(Indonesia)       | Light yellow             | Firm                       | Long, tapered    | Mild, pleasant<br>flavour, less<br>aroma  | Late, September-<br>October |

Source: Horticulture Sciences Department, Florida Cooperative Extension Service Institute of Food and Agricultural Sciences, University of Florida.

## 2.2. Jackfruit seed

Jackfruit seeds are substantial by-products that account for more than 15% of a fruit's total weight (Prathima, 2008). According to early research, this section of the jackfruit is a good source of vital nutrients like carbohydrates, protein, and minerals (Ocloo *et al.*, 2010). Jackfruit seed's market has expanded as consumer insight of the diet-disease association has grown. It is thought to be a valuable functional food element because it provides additional physiological benefits (Waghmare *et al.*, 2019). Although boiled or roasted jackfruit seeds are commonly consumed, the majority of jackfruit seeds are thrown (Zhang *et al.*, 2016). Many scientists have looked into jackfruit seeds as a potential supply of starch (Mahanta *et al.*, 2015). The seed contains around 20% (dry basis) and as per Tulyathan *et al.*, 2002, the recovered yield of starch extracted from jackfruit seeds was about 77%, implying its potential use as a potent source of starch in the food and pharmaceutical industries (Patana and Patana, 2011) as a stabiliser, thickener and binding agent (Rengsutthi and Charoenrein, 2011). Converting fresh seeds to flour is a good technique to increase their use (Mahanta and Kalita, 2015). Jackfruit seed also comprises a significant level of non-reducing carbohydrates, making it an ideal candidate for use as a prebiotic (Bhornsmithikun *et al.*, 2010), because of its high protein content, jackfruit seeds have also been explored as a raw material for ethanol production, which can be regarded as a renewable source of energy (Nuriana and Wuryantoro, 2015) and as a protein source in the food sector (Vazquez *et al.*, 2017). Roasted Jackfruit seeds can be powdered and utilised to add value to a variety of goods to increase their shelf-life. By combining it along with wheat flour and different varieties of low-priced flours, jackfruit seed powder is utilised as a substitute flour in bakery and confectionery items (Hossain, 2014). Banerjee and Datta (2015) reported that the seeds are served by boiling or roasting them in certain parts of India, or they are used to augment potato.

Malnutrition is a prominent issue in India, owing to a lack of protein intake. To combat malnutrition, jackfruit seeds could be employed as a cost-effective alternative protein source (Chowdhury *et al.*, 2012). Jackfruit seed's market has expanded as consumer insight of the diet-disease association has grown. It is thought to be a valuable functional food element because it provides additional physiological benefits (Waghmare *et al.*, 2019).

### 2.3. Chemical composition and nutritive value of jackfruit and jackfruit seed

Depending on the kind, jackfruits have different chemical compositions. Protein, calcium, iron and thiamine content is abundant in jackfruit flesh and seeds than other tropical fruits (Haq *et al.*, 2006).

**Table 2.2. Nutritional profile of jackfruit (Value per 100 g edible portion)**

|                           |             |
|---------------------------|-------------|
| <b>Proximates</b>         |             |
| Carbohydrates (g)         | 16.0 - 25.4 |
| Protein (g)               | 1.2 - 1.9   |
| Fat (g)                   | 0.1 - 0.4   |
| Dietary Fibre (g)         | 1.0 - 1.5   |
| <b>Vitamins</b>           |             |
| Vitamin A (IU)            | 175 - 540   |
| Thiamine (mg)             | 0.03 - 0.09 |
| Riboflavin (mg)           | 0.05-0.4    |
| Niacin (mg)               | 0.92        |
| Pantothenic acid (mg)     | 0.235       |
| Vitamin B6 (mg)           | 0.329       |
| Folate (µg)               | 24.0        |
| Vitamin C (mg)            | 7.0 - 10.0  |
| <b>Water and Minerals</b> |             |
| Calcium (mg)              | 20.0 - 37.0 |

|                  |               |
|------------------|---------------|
| Iron (mg)        | 0.5 - 1.1     |
| Magnesium (mg)   | 27.0          |
| Phosphorous (mg) | 38.0 - 41.0   |
| Potassium (mg)   | 191.0 - 407.0 |
| Sodium (mg)      | 2.0 - 41.0    |
| Zinc (mg)        | 0.13          |
| Water (g)        | 72.0 - 94.0   |

Sources: Adapted from USDA, United States Department of Agriculture report (2016, May); Ranasinghe *et al.* (2019).

From Table 2.2, it is evident that jackfruit is a good source of potassium with (448 mg) per 100g along with this it provides ample supply of thiamine, niacin, calcium, sodium and vitamin B6. Jackfruit is rich in protein content with (1.72g) proteins in 100g compared to other tropical fruits such as banana (1.09g), mango (0.82g), fig (0.75g) and pineapple (0.54g), as reported by United States Department of Agriculture (2016).

**Table 2.3. Phytochemical content of jackfruit pulp and seed**

| Phytochemical | Jackfruit Pulp (mg / 100 g) | Jackfruit Seed (mg / 100 g) |
|---------------|-----------------------------|-----------------------------|
| Phytic Acid   | 6.14 ± 0.08                 | 8.11 ± 0.06                 |
| Oxalate       | 6.39 ± 0.13                 | 5.53 ± 0.06                 |
| Alkaloid      | 7.88 ± 0.06                 | 8.85 ± 0.08                 |
| Tannin        | 0.03 ± 0.01                 | 0.06 ± 0.01                 |
| Flavonoid     | 3.91 ± 0.08                 | 2.03 ± 0.06                 |

The (±) values are the mean standard deviation of duplicate analysis.

Source: Amadi *et al.*, 2018

Studies conducted by Arung *et al.* (2007), Chandrika *et al.* (2004), Lin and Lu (1993), Ong *et al.* (2006), Venkataraman (2001), and Wong *et al.* (1992) reports that a variety of phytochemicals, including carotenoids, flavonoids, volatile acids, sterols, and tannins are present in jackfruit, in varying concentrations depending on the variety. The cultivar type and the habitat affects the phytochemical components in jackfruit (Haq, 2006; Baliga *et al.*, 2011). It has already been documented that the composition of jackfruit seeds is similar to that of grains (Rahman *et al.*, 1995). These fruit seeds can be used to isolate various triterpenes and flavonoids. Other than triterpenes and flavonoids these seeds also contain alkaloids, phenols, tannins, saponins, steroids etc. (Swami *et al.*, 2012). Table 2.3 shows the phytochemical makeup of jackfruit seed and pulp. The level of phytic acid ranged from 6.14 mg/100g (pulp) to 8.11 mg/100g (seed), oxalate from 3.69 mg/100g (pulp) to 5.53 mg/100g (seed), and alkaloid from 7.88 mg/100g (pulp) to 8.85 mg/100g (seed). The tannin concentration of jackfruit seed (0.06 mg/100g) was measured, but that of the pulp (0.03 mg/100g) was lower. The seed has 2.03 mg/100g of flavonoid content, while the pulp has 3.91 mg/100g.

#### **2.4. Food applications of jackfruit seed**

Due to their antinutrient composition, jackfruit seeds should not be eaten directly. Two of the most common cooking methods are boiling and roasting.

To add extra flavour and crunchy texture to various dishes jackfruit seeds can be used. They are large in size and can be used in different ways (Bentilan, 2019) for instance seeds from jackfruit have been utilised to make a variety of value-added products. The bread industry in India is one of the most quickly growing organised food businesses. Baked products are growing more popular since they are easy to find, inexpensive, available in a variety of flavours and textures, and are high in nutritional value. Their key advantage is the ease with which bread products can be augmented with useful substances. As a result, bakery products are an effective vehicle for delivering potent nutrients that benefit consumer's health (Waghmare *et al.*, 2019).

When jackfruit seed flour was added to a variety of baked cereal goods, the nutritional value increased. Arpit and John (2015) investigated the effect of increasing jackfruit seed flour supplementation on the quality attributes of chocolate cake by changing the wheat flour and jackfruit seed flour proposals. When 10g of jackfruit seed flour was added to 100g of wheat flour used to make chocolate cake, the protein and ash levels went up while the fat content went down.

The amount of ash and crude fibre in biscuits made with jackfruit seed flour increased. The amount of minerals in flour is related to the amount of ash in the flour. Since jackfruit seeds are high in crude fibre, they increase the total amount of crude fibre in the biscuits while decreasing the protein and carbohydrate content.

The crude fibre level of bread made with jackfruit seed flour was higher than that of other baked goods, which are often lacking in dietary fibre (Islam *et al.*, 2015). The colour of the bread stayed unchanged when it was reinforced with a 25% substitution of jackfruit seed flour, however it changed from light brown to dark brown when more jackfruit seed flour was added. Likewise, the control sample and the sample with 25% seed flour were equally favoured in terms of flavour. The control sample and the one augmented with 25% jackfruit seed flour tasted similar, and the texture of the bread with 25% seed flour was favoured (Hossain, 2014).

## **2.5. Health benefits of jackfruit and its seeds**

Because of their high fibre content, the seeds can lessen the risk of heart disease, prevent constipation, and encourage weight loss. Resistant starch, found in jackfruit seeds, helps to regulate blood sugar and keep the gut healthy. Jackfruit seeds have antimicrobial activity, which helps to prevent foodborne illnesses (Maurya & Mogra, 2016), and they contain a lectin called jacalin, which is used to assess an HIV-positive person's immune system. The seeds are known to help overcome the toxicity caused by alcohol in China, and they are also a key component of an antidote made for heavy drinkers in India (Butool & Butool, 2013). The seeds are a good source of magnesium, which is important for decreasing blood pressure and supporting bone health since it increases calcium absorption and hence helps to strengthen bones (Maurya & Mogra, 2016). The seeds are also abundant in highly soluble protein, which helps to prevent and relieve mental stress and anxiety. The seeds have a low capacity for absorbing water and fat, which aids in the prevention of obesity (Waghmare *et al.*, 2019).

## **2.6. Health benefits and nutritional value of traditional hummus**

Several studies have linked eating hummus/chickpea to weight loss, glucose and insulin response, cardiovascular disease, cancer, and/or gastrointestinal health (Wallace *et al.*, 2016). Diets high in fibre, low in calorie density and glycemic load, and moderate in protein are

considered to be extremely advantageous for weight management (Albete *et al.*, 2010). In a study of 10 healthy subjects consuming hummus, it showed four times lower postprandial glucose responses than that of white bread (Augustin *et al.*, 2016). Soluble fibre is widely documented for its beneficial effects on TC and LDL-C, two confirmed biomarkers for cardiovascular disease (National Academies Press: Washington, DC, USA, 2005). Consumption of a chickpea diet (200 g/day) produces butyrate, a principal short chain fatty acid in healthy adults (Fernando *et al.*, 2010). Suppression of cell proliferation and inducing of apoptosis, are two widely reported functions of butyrate; this may help in reduction of colorectal cancers (Cummings *et al.*, 1981). Lycopene, biochanin A, and saponins are among several other dietary bioactive compounds found in chickpeas and hummus, and are reported to reduce the risk of certain types of cancers (Mathers *et al.*, 2002). Soluble and insoluble fibres are two types of dietary fibre. Insoluble fibres are indigestible and encourage bowel movements, whereas soluble fibres are slowly digested in the colon. When chickpeas and/or hummus are introduced to the diet, marked increases in dietary fibre intake have been recorded, similar to other plant foods, legumes, and pulses (O'Neil *et al.*, 2014; Murty *et al.*, 2010; Nestel *et al.*, 2004). When compared to a regular diet, human studies using chickpeas show overall advantages in bowel health, including increased frequency of defecation, ease of defecation, and softer stool composition (Murty *et al.*, 2010; Nestel *et al.*, 2004). As per U.S. Food and Drug Administration, because of its particular blend of health-promoting ingredients, traditional hummus may help consumers maximise their nutrient-to-calorie ratio when picking a dip or spread.

Chickpeas and tahini are the two main ingredients of hummus, both of which are high in important nutrients and bioactive microconstituents. Chickpeas are pulses, edible seeds of plants in the legume family. Protein, fibre, resistant starch, and unsaturated fatty acids, a variety of vitamins and minerals, including riboflavin, niacin, thiamin, folate, potassium, phosphorus, and calcium are present in chickpeas (Dadon *et al.*, 2017; Jukanti *et al.*, 2012). Tahini is a paste made from toasted ground sesame seeds which contain unsaturated fatty acids, antioxidant lignans and tocopherols, and important minerals such as calcium and phosphorus (Prasad *et al.*, 2012).

**Table 2.4. Nutritional profile of traditional hummus**

| Nutrient              | Value per 100g |
|-----------------------|----------------|
| Energy (Kcal)         | 166            |
| Carbohydrates (g)     | 14.29          |
| Protein (g)           | 7.90           |
| Fat (g)               | 9.60           |
| Dietary Fibre (g)     | 6.0            |
| Thiamine (mg)         | 0.180          |
| Riboflavin (mg)       | 0.064          |
| Niacin (mg)           | 0.528          |
| Pantothenic acid (mg) | 0.132          |
| Vitamin B6 (mg)       | 0.200          |
| Folate (µg)           | 83             |
| Vitamin C (mg)        | 0              |
| Calcium (mg)          | 38             |
| Iron (mg)             | 2.44           |
| Magnesium (mg)        | 71             |
| Phosphorous (mg)      | 176            |
| Potassium (mg)        | 228            |
| Sodium (mg)           | 379            |
| Zinc (mg)             | 1.83           |

Source: USDA National Nutrient Database for Standard Reference, Nutrient Database Number (NDB No.) in the USDA Food Composition Database. (<https://ndb.nal.usda.gov/ndb/>)

Hummus is a good source of plant-based protein, as seen in (Table 2.4). It is a moderate-carbohydrate content food product that contains key vitamins and minerals such as calcium, iron, phosphorus, potassium, and B vitamins in large proportions.

## 2.7. Comparison between chickpea and jackfruit seed

**Table 2.5. Proximate nutritional value of chickpea and jackfruit seed (%)**

| Proximate nutrient | Chickpea      | Jackfruit seed |
|--------------------|---------------|----------------|
| Moisture           | 5.62 - 8.17   | 71.59 - 72.25  |
| Carbohydrate       | 37.2 - 50.4   | 7.76 - 8.02    |
| Protein            | 19.68 - 22.75 | 9.98 - 10.20   |
| Fat                | 4.18 - 4.92   | 4.17 - 4.41    |
| Crude Fibre        | 0.26 - 1.43   | 3.86 - 3.98    |

Source: Benu Singhai and S. K. Shrivastava, 2006; Amadi *et al.*, 2018

The above table (Table 2.5) shows comparison between the proximate nutrients of chickpea and jackfruit seed. It is evident that the jackfruit seed is very significantly high in moisture, but the case is vice-versa when it comes to carbohydrates. The seed has a good amount of protein, but this amount is low in comparison to chickpea. Both chickpea and jackfruit seed have a similar fat profile. Crude fibre is notably more in the seed. Less carbohydrate, and more crude fibre makes the seed a better alternative for individuals suffering from diabetes and gastrointestinal disorders

**Table 2.6. Mineral content of chickpea and jackfruit seed (Value per 100g)**

| <b>Mineral</b> | <b>Chickpea</b> | <b>Jackfruit seed</b> |
|----------------|-----------------|-----------------------|
| Calcium (mg)   | 57              | 40                    |
| Potassium (mg) | 718             | 220                   |
| Manganese (mg) | 0.656           | 9.50                  |
| Iron (mg)      | 4.31            | 18.25                 |
| Zinc (mg)      | 2.76            | 9.28                  |

Source: USDA National Nutrient Database for Standard Reference, Nutrient Database Number (NDB No.) in the USDA Food Composition Databases. (<https://ndb.nal.usda.gov/ndb/>); Amadi *et al.*, 2018.

The table (Table 2.6) given above compares certain minerals between chickpea and jackfruit seed. Jackfruit seed shows a remarkable drop in the level of potassium and calcium. On the other hand, it has appreciably high amounts of manganese, iron, and zinc. Therefore, it can be said that jackfruit seed serves as a balanced mineral source making it a good alternative for chickpea.

## CHAPTER 3 - MATERIALS AND METHODOLOGY

### 3.1. Sample collection



Soft variety Jackfruit (*Artocarpus heterophyllus* Lam.) was bought from Dindigul, Tamil Nadu. 46 seeds from 1 kg pulp of the same were procured.

White sesame seeds, olive oil, garlic, cumin powder, chilli flakes, lemon, salt, and 500 mL water were bought from Nilgiris supermarket, Padur, Tamil Nadu.

### 3.2. Formulation of jackfruit seed hummus

Selection of good quality ripen jackfruit

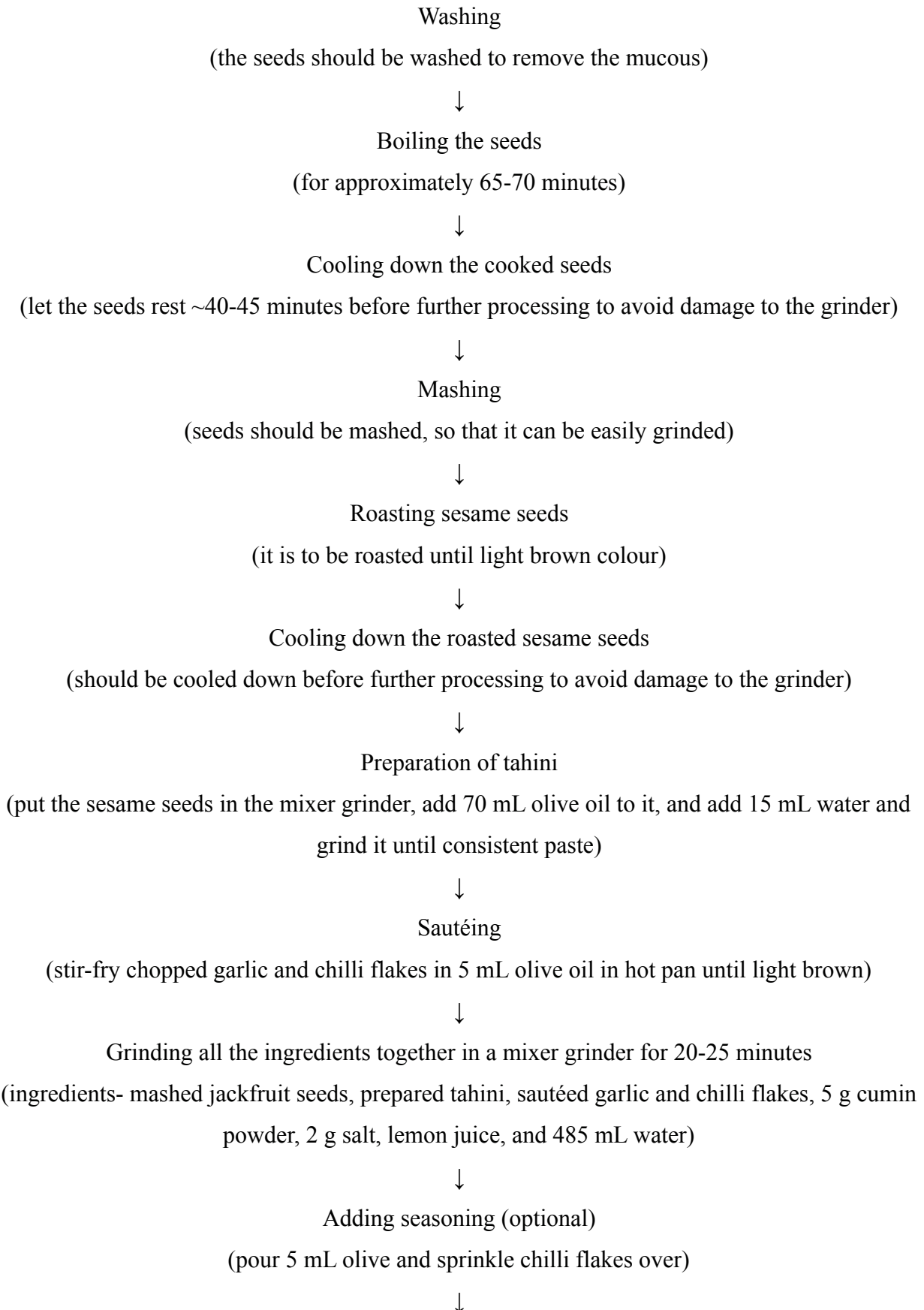


Procuring the pulps  
(using sterile knife)



Removal of seeds from the procured pulp  
(manually using sterile knife)





Jackfruit Seed Hummus  
(smooth-textured paste called hummus is obtained)

**Figure 3.1. Flowchart for formulation of jackfruit seed hummus**



Jackfruit pulp



Seeds



Washing the seeds



Boiling the seeds  
(for ~ 65-70 minutes)



Peeling the seeds



Mashing the boiled, peeled seeds



Roasting of white sesame seeds  
(until light brown)





Preparation of tahini  
(Roasted sesame seeds + 70 mL olive oil)



Sautéing chopped garlic cloves and chilli flakes in 5 mL olive oil





Grinding all the ingredients together in a mixer grinder for 20-25 minutes  
(ingredients- mashed jackfruit seeds, prepared tahini, sautéed garlic and chilli flakes, 5 g cumin powder, 2 g salt, lemon juice, and 485 mL water)



Jackfruit Seed Hummus  
(with seasoning- 5mL olive oil, chilli flakes)

**Figure 3.2. Flowchart for formulation of jackfruit seed hummus (Pictorial representation)**

**Table 3.1. Formulation of jackfruit seed hummus**

| S. No. | Ingredients                         | Quantity (in g) |
|--------|-------------------------------------|-----------------|
| 1.     | Jackfruit Seeds (boiled and peeled) | 469.72          |
| 2.     | White Sesame Seeds                  | 27.10           |
| 3.     | Olive Oil (80 mL)                   | 71.68           |
| 4.     | Garlic Cloves                       | 26.70           |
| 5.     | Cumin Powder                        | 5.0             |
| 6.     | Red Chilli Flakes                   | 2.0             |
| 7.     | Lemon Juice (4 mL)                  | 3.89            |
| 8.     | Water (500 mL)                      | 508.10          |

### 3.3. Sensory evaluation

Sensory evaluation was conducted for the freshly-made jackfruit seed hummus by the panellists to evaluate the sensory characteristics of the sample. The sensory test was conducted in a laboratory under normal (daylight) illumination and ambient temperature. Panellists evaluated the sensory properties of the hummus samples based on their degree of like. The 9 point hedonic scale was used to evaluate appearance, taste, texture, aroma and overall acceptability, where one (1) was “dislike extremely”, two (2) was “dislike very much”, three (3) was “dislike moderately”, four (4) was “dislike slightly”, five (5) “neither like nor dislike”, six (6) was “like slightly”, seven (7) was “like moderately”, eight (8) was “like very much”, nine (9) was “like extremely”. In addition an acceptance rate (%) was calculated for the developed hummus samples.

**Table 3.2. Sensory evaluation panel**

| S. No. | Criterion             | Hedonic Scale Point (Avg.) |
|--------|-----------------------|----------------------------|
| a.     | Appearance            |                            |
| b.     | Aroma                 |                            |
| c.     | Taste                 |                            |
| d.     | Texture               |                            |
| e.     | Overall Acceptability |                            |

### **3.4. Characterization of the formulated jackfruit seed hummus**

#### **3.4.1. Proximate Analysis (formulated hummus)**

##### **3.4.1.1. Moisture content (Hot-air oven method)**

###### **Apparatus**

- Moisture dish – Porcelain, silica, glass or Aluminium (7.5 x 2.5 cm)
- Oven: electric maintained at  $105 \pm 10^{\circ}\text{C}$
- Desiccator

###### **Procedure**

About 5 gm of the sample was weighed accurately in the moisture dish previously dried in an oven and weighed. The dish was placed in the oven maintained at  $105 \pm 10^{\circ}\text{C}$  for 4 hours. It was then cooled in the desiccator and weighed. The process of drying, cooling, and weighing was repeated at a 30 minutes interval until the difference in two consecutive weights was less than 1 mg. The lowest weight was recorded.

Moisture percentage was calculated according to the formula:

$$\text{Moisture (\%)} = \frac{(W1 - W2)}{(W1 - W)} \times 100$$

where

W= mass in g, of the empty dish;

W1 = mass in g, of the dish with the material before drying; and

W2 = mass in g, of the dish with the material after drying to constant mass.

### **3.4.1.2. Total ash content**

#### **Procedure**

About 5 g of the preserved material weighed accurately in a tared, clean and dry porcelain or silica dish. The material in the dish was ignited with the flame of a suitable burner for about 1 hour. The ignition was completed by keeping in a muffle furnace at  $550 \pm 10^{\circ}\text{C}$  until gray ash resulted. It was then cooled in a desiccator and weighed. The process of igniting, cooling and weighing was repeated at half-hour intervals until the difference between two successive weighings was less than 1 mg. The lowest mass was noted.

#### **Calculation**

$$\text{Total ash (on dry basis), percent by mass} = \frac{(M_2 - M)}{(M_1 - M)} \times 100$$

where,

M = mass in g of the empty dish;

M1 = mass in g of the dish with the dried material taken for the test; and

M2 = mass in g of the dish with the ash.

### **3.4.1.3. Protein estimation (Kjeldahl method)**

Protein content was determined using the micro kjeldahl method as described by AOAC (2005). 2 grams of the sample was put into the digestion flask along with copper sulphate and sodium sulphate (catalyst) in the ratio 5:1 respectively and 25 ml concentrated sulphuric acid were also added to the digestion flask. The flask was placed into the digestion block in the fume cupboard and heated until frothing ceased giving clear and light blue green coloration the mixture was then allowed to cool and diluted with distilled water until it reached 250 ml of volume flask.

Distillation apparatus was connected and 10 ml of 40 % sodium hydroxide was added. The released ammonia by boric hydroxide acid was then treated with 0.02 m of hydrochloric acid until the green colour changed to purple. Percentages of nitrogen in the sample was calculated using the formulated below:

$$\% \text{ nitrogen} = \frac{\text{Litre} - \text{Blank} \times \text{normality}}{\text{Weight of sample}} \times 100$$

$$\% \text{ crude protein} = \% \text{ nitrogen} \times 6.25$$

#### **3.4.1.4. Crude fat estimation (Soxhlet method)**

The crude fat content of the hummus samples were determined using Soxhlet method (Soxtec Model-8000; AOAC 2000, 4.5.01). The flasks used for the extraction were cleaned by placing them in a drying oven at 92°C for 1 h and cooled in desiccators. The mass of the cooled flasks were measured by analytical balance (M1). About 2 g of the sample was weighed into each of the thimbles lined with cotton at their bottom. The thimbles with the sample content were placed into the Soxtec soxhlet extraction apparatus. A 70 ml of diethyl ether was added into each flask used for the extraction. The extraction process was done for about 4 hrs and then after the flasks with the remaining contents were removed from the soxhlet and placed in a drying oven at 92°C for 1 hour. The flasks were then placed in desiccators for 30 min. The masses of each flask together with its fat contents was measured (M2).

The crude fat content was calculated from the equation:

$$\% \text{ crude fat} = \frac{(M2 - M1)}{W} \times 100$$

where,

M2 = mass of flask and fat extracted,

M1 = mass of dried flask,

W = sample weight.

#### **3.4.1.5. Crude fibre estimation**

The crude fibre was determined using the methods by (AOAC, 2005). About 5 g of each sample was weighed into a 500 ml Erlenmeyer flask and 100 ml of TCA digestion reagent was added. It was then brought to boiling and refluxed for exactly 40 minutes counting from the start of boiling. The flask was removed from a 15:0 cm number 4 whatman paper. The residue was washed with hot water stirred once with a spatula and transferred to a porcelain dish. The sample was dried overnight at 105 °C. After drying it was transferred to a desiccator and weighed as W1. It was then burnt in a muffle furnace at 500°C for six (6) hours, allowed to cool and then re-evaluated as W2.

$$\% \text{ Crude fibre} = \frac{W1 - W2}{W0} \times 100$$

where,

W1 = Weight of crucible + fibre + ash

W2 = Weight of crucible + ash

W0 = Dry weight of food sample

#### **3.4.1.6. Carbohydrate content**

The carbohydrate content of each sample was determined using the method described by (AOAC, 2005). This was determined by the differences using the formulation calculated by using the formula below.

$$\% \text{ carbohydrate} = 100 - (\% \text{ moisture} + \% \text{ ash} + \% \text{ protein} + \% \text{ crude fat} + \% \text{ crude fibre})$$

#### **3.5.2. Energy**

The energy values (Kcal/100 g samples) of the jackfruit seed hummus were estimated using the factors for fat (9 Kcal/g), protein (4 Kcal/g), and carbohydrate (4 Kcal/g).

$$\text{Energy (Kcal/100g)} = (\text{Fat \%} \times 9) + (\text{Protein \%} \times 4) + (\text{Carbohydrate \%} \times 4)$$

### 3.5.3. Phyto-chemical Analysis

#### 3.5.3.1. Alkaloids (boiled jackfruit seed)

5.09 g of the sample was weighed and dispersed into 50 mL of 10% acetic acid solution in ethanol. The mixture was well-shaken and then allowed to stand for about 4 hours. The mixture was then filtered. The filtrate was evaporated to one quarter of its original volume on the hot plate. Concentrated ammonium hydroxide was added dropwise in order to precipitate the alkaloids. A pre-weighed filter paper was used to filter the precipitate, and it was then washed with 1% ammonium hydroxide solution. The filtrate containing the precipitate dried at 60°C for 30 minutes, transferred into dessicator to cool, and then re-weighed until a constant weight was obtained.

$$\text{Alkaloid content (g)} = \frac{(\text{After dry weight} - \text{Empty weight})}{\text{Sample weight}} \times 100$$

#### 3.5.3.2. Oxalate (boiled jackfruit seed)

The sample was dried at 70°C at 25 mm in a vacuum oven for 24 hours. Now, take some amount of sample in a mortar and pestle and grind it as fine powder, and pass it through a 0.5 mm mesh. Weigh accurately 1.0 g of sample and dissolve in 100 mL of 0.75 M H<sub>2</sub>SO<sub>4</sub>. The solution was then carefully stirred with a magnetic stirrer for 1 hour and filtered. 25 mL of the filtrate was pipetted and titrated hot (80°C - 90°C) against 0.1 M KMnO<sub>4</sub> to an end point of a faint pink colour that persisted for more than 30 seconds.

$$\text{Oxalate content} = \text{Titre value} \times \text{constant (0.225)}$$

#### 3.5.3.3. Total flavonoids (formulated hummus)

The total flavonoid content (mg/mL) was determined using aluminium chloride (AlCl<sub>3</sub>) method. Weigh 1.0056 g of samples were weighed and 10 mL 60% alcohol (aqueous) was used to extract flavonoids. These samples were kept in ultrasonics for 30 minutes. The samples were then

centrifuged at 3000 g. All the supernatants were transferred to 25 mL volumetric flask and then was fixed to 25 mL with 60% aqueous ethanol. 1.5 mL each extract and 4.5 mL of distilled water pipetted into a 25 mL test tube and then mixed with 1mL 5% NaNO<sub>2</sub>. It was then kept for 60 minutes for incubation. 1 mL of 10% Al(NO<sub>3</sub>)<sub>3</sub> solution was added to the mixture. The mixture was kept for 6 minutes before adding 10 mL 4% NaOH solution and fixed to 25 mL with 60% aqueous ethanol. Finally, the mixture was reacted for 15 minutes and the absorbance of the solution was measured with a spectrophotometer at 510 nm.

(Blank- 5 mL of the extraction solvent)

## CHAPTER 4 - RESULTS AND DISCUSSION

### 4.1. Jackfruit Seed

#### 4.1.1. Phyto-chemicals

**Table 4.1. Alkaloid and oxalate content in boiled jackfruit seed**

| Phyto-chemical | Boiled jackfruit seed (mg / 100 g) |
|----------------|------------------------------------|
| Alkaloid       | 1.05                               |
| Oxalate        | 0.05                               |

**Table 4.2. Alkaloid and oxalate content in raw jackfruit seed**

| Phyto-chemical | Raw jackfruit seed (mg / 100 g) |
|----------------|---------------------------------|
| Alkaloid       | 8.77 - 8.93                     |
| Oxalate        | 5.47 - 5.59                     |

Source: Amadi *et al.*, 2018.

Alkaloids and oxalates are anti-nutritional factors that have evident physiological effects on humans. Alkaloids cause neurological disorders either by disrupting or by augmenting electrochemical transmission. Its consumption might also result in rapid heartbeat, paralysis and death in extreme cases. Oxalates hinder calcium and magnesium metabolism and form complexes with proteins. They have antagonistic effects on peptic digestion, thus causing gastrointestinal problems. Antinutritional factors can be effectively reduced by almost all thermal processing treatments. They show higher susceptibility to moist heat than dry heat. Soaking, germination, boiling, autoclaving, fermentation, genetic manipulation and other processing methods could remove most of the toxic and anti-nutrient effects of these compounds. (Thakur *et al.*, 2018).

It can be seen in Table 4.4 that boiled jackfruit has 1.05 mg/100g alkaloid and 0.05 mg/100g

oxalate content. This is a very low amount of these anti-nutritional factors, thus the seed can be considered safe for human consumption. But as per the secondary data collected (Table 4.2), it was found that raw jackfruit seed contains high alkaloid (8.77 - 8.93 mg/100g) and oxalate (5.47 - 5.59 mg/100g) content. Therefore, it can be concluded that boiling has significantly reduced the alkaloid and oxalate content of the jackfruit seed, hence, safe to consume.

## **4.2. Jackfruit Seed Hummus**

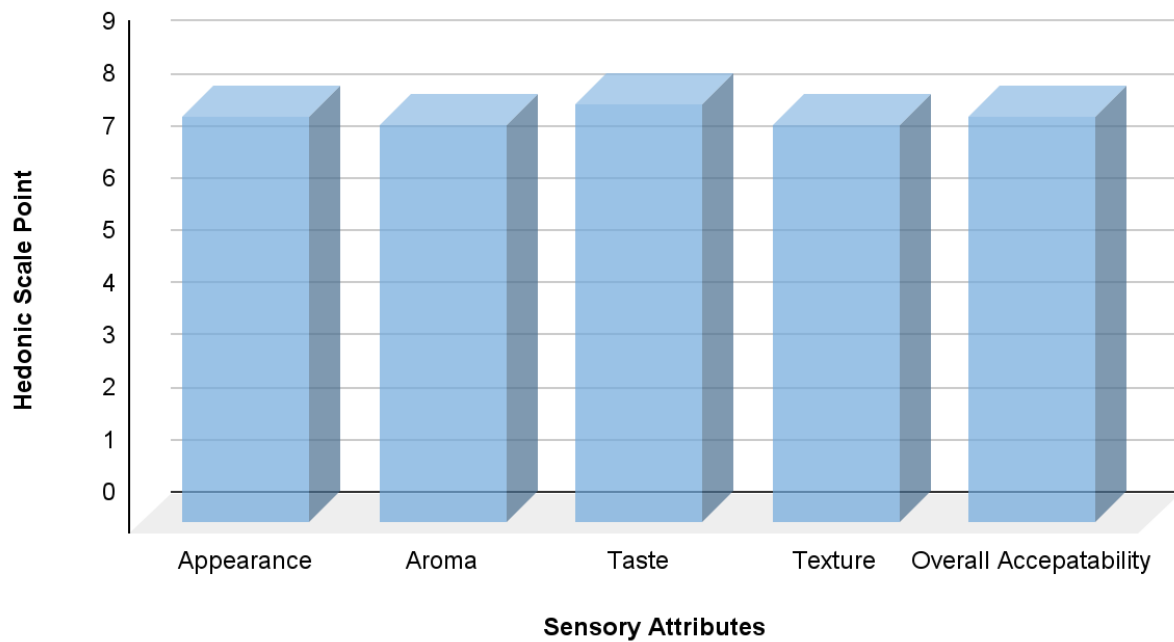
### **4.2.1. Sensory evaluation**

Sensory evaluation was conducted for 10 people of different ages. The average age of the test population was 28 years. To get the final result, average hedonic points were calculated for each of the given five sensory attributes, i.e., appearance, aroma, taste, texture, and overall acceptability. The average hedonic scale points are mentioned in the table and the graph below:

**Table 4.3. Sensory evaluation panel**

| <b>S. No.</b> | <b>Criterion</b>      | <b>Hedonic Scale Point (Avg.)</b> |
|---------------|-----------------------|-----------------------------------|
| <b>a.</b>     | Appearance            | 7.75                              |
| <b>b.</b>     | Aroma                 | 7.625                             |
| <b>c.</b>     | Taste                 | 8                                 |
| <b>d.</b>     | Texture               | 7.625                             |
| <b>e.</b>     | Overall Acceptability | 7.75                              |

**Average Hedonic Scale Points**



It can be observed in the graph that the attribute, 'taste' peaks through the y-axis with a point 8, i.e., liked very much, on a 9-point hedonic scale. Appearance, aroma, texture, and overall acceptability were above 7 (liked moderately), but fell a little below 8 (liked very much). It shows that the jackfruit seed hummus is a decently acceptable product.

#### 4.2.2. Nutritional profile of jackfruit seed hummus

**Table 4.4. Nutritional profile of Jackfruit seed hummus**

| <b>Nutrient</b>      | <b>Values (g/100 g)</b> |
|----------------------|-------------------------|
| Moisture             | 18.36                   |
| Total Ash            | 0.82                    |
| Carbohydrate         | 74.59                   |
| Protein              | 4.79                    |
| Fat                  | 1.32                    |
| Fibre                | 0.12                    |
| <b>Energy (Kcal)</b> | <b>329.4</b>            |

**Table 4.5. Nutritional profile of Chickpea hummus**

| <b>Nutrient</b>      | <b>Value (g/100g)</b> |
|----------------------|-----------------------|
| Carbohydrates        | 14.29 - 15            |
| Protein              | 7.78 - 7.90           |
| Fat                  | 9.60 - 17.82          |
| Fibre                | 5.5 - 6.0             |
| <b>Energy (Kcal)</b> | <b>166 - 237</b>      |

Source: Wallace *et al.* (2016) and Reister *et al.* (2020).

Table 4.4 shows nutritional values of the formulated jackfruit seed hummus. The energy, moisture, and carbohydrate content were reported high, whereas mineral, protein, fat, and fibre were less. In contrast to the secondary data on chickpea hummus (Table 4.5), jackfruit seed hummus tends to be a high-carb, low-fat food.

In the American Diabetes Prevention Program and the Finnish Diabetes Prevention Trial, it was exemplified that over a 4-year study period, low-fat, high-carbohydrate diets when combined with physical activity and behavioral strategies, maintained modest weight loss (3–4 kg) with a marked reduction in the risk of developing type 2 diabetes mellitus. Therefore, it may have a role in weight maintenance. (Uchegbu and Kopelman, 2013). A study conducted by the National Institutes of Health found that there was a greater fat loss from a high-carb, low-fat diet when compared to plans that restricted carbohydrate consumption. The study was conducted for obese individuals who received the same amount of calories from a diet that was either high-carb and low-fat or high-fat and low-carb. Those on a high-carbohydrate diet lost more body fat than the people with restricted carbohydrate intake. (Hall *et al.*, 2015).

Thus, it can be said that jackfruit seed hummus can assist weight loss and may even contribute to weight maintenance.

#### 4.2.3. Phyto-chemicals

**Table 4.6. Total flavonoid content in jackfruit seed hummus**

| Phyto-chemical   | Jackfruit seed hummus (mg/100 g) |
|------------------|----------------------------------|
| Total flavonoids | 1.16                             |

**Table 4.. Total flavonoid content in raw jackfruit seed**

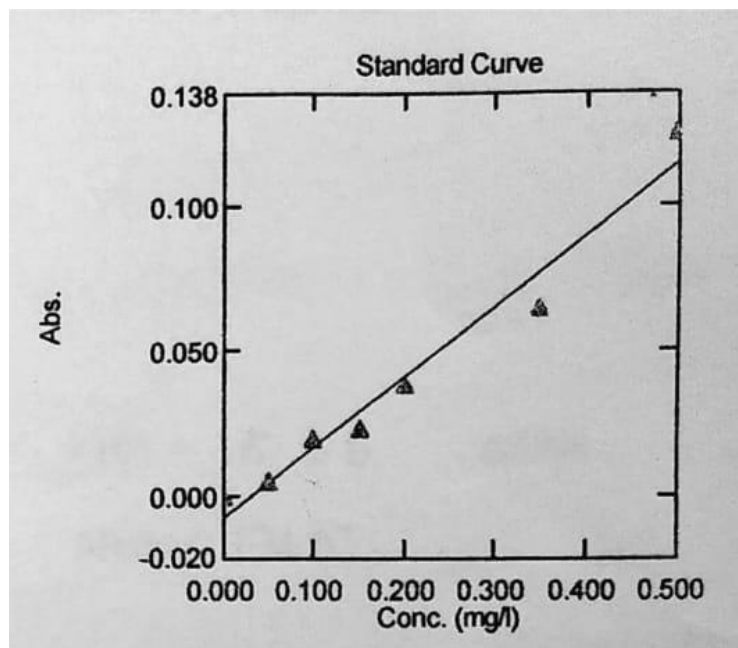
| Phyto-chemical   | Raw jackfruit seed (mg/100 g) |
|------------------|-------------------------------|
| Total flavonoids | 1.97 - 2.09                   |

Source: Amadi *et al.*, 2018.

Table 4.6 shows total flavonoid content of the formulated hummus to be 1.16 mg/100g. Flavonoids are plant secondary metabolites that are claimed to have natural antioxidant molecules (Ioannou et al., 2020). They are said to be heat-sensitive phyto-compounds; but, their sensitivity to heat depends upon their structure (Chaaban *et al.*, 2017).

In contrast to the secondary data on total flavonoid content of raw jackfruit seed (Table 4.7), it is

observed that the processing done for hummus formulation did not have significant effect on the total flavonoid content of the seed.



**Graph 4.1. Total flavonoids- Concentration of the aqueous extract of the sample vs. Absorbance**

As observed from the graph, when the concentration of the sample increased, the absorbance increased as well; the concentration of the sample shows direct proportionality with the absorbance. Maximum absorbance was noted on 0.500 mg/L of concentration.

## **CHAPTER 5 - CONCLUSION**

Jackfruit is mostly grown in smallholder farms and produces a variety of goods for food, feed and industry. Since downstream goods have been limited, jackfruit has remained mostly a minor tropical fruit in domestic trade. Despite the fact that jackfruit is vital for these objectives, no tree enhancement research has been done on significant yet overlooked fruit species. During the fruiting season, the jackfruit tree produces a huge number of fruits and people use them in a variety of ways, yet several fruits and seeds are discarded on a massive scale due to its year-around availability. They are mostly grown in householders' farms between the month of May-October and which makes it easily available and economically low-cost than chickpea. As a result, raising awareness about the functional and therapeutic applications of jackfruit is vital. Considering jackfruit seed's abundance, plethora of advantages, prospects for global expansion, and popularity of chickpea hummus, formulation of jackfruit seed hummus was well-thought of as an effective and innovative way of food waste management. It was seen during the course of this project that both jackfruit seed and chickpea have similar taste and texture when boiled. Acknowledging the above set of information, jackfruit seeds were selected to substitute chickpea. Through the results it can be concluded that jackfruit seed can effectively substitute chickpea in hummus with a decently accepted score of 8 out of 9 in terms of taste, and not less than 7.5 out of 9 in terms of other sensory attributes. The formulated hummus emerged as a high-carb, low-fat food, making it one of the dietary treatments for obesity, and good for weight maintenance when combined with physical activity and behavioral strategies.

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